

The Plant's Secret Wardrobe

I PUC · Ch 5 · Morphology of Flowering Plants

Starring Bindu 👧 & Prof. BAIO 🦉

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Meet Mr Mango

A friendly mango tree greeted Bindu. "**Morphology** is the study of my outer form. I have **roots** below, a **stem** above, then **leaves**, **flowers**, **fruits** and **seeds**. But many plants dress up in disguise!"

💡 **REMEMBER**

Dicots usually have a tap root system; monocots have a fibrous root system.



Roots in Disguise

Carrot and radish swelled up, storing food. The banyan dropped **prop roots** to hold up its heavy branches. Maize and sugarcane stood on **stilt roots**. In a swamp, Rhizophora pushed **pneumatophores** up into the air to breathe.

💡 **REMEMBER**

Prop roots — banyan · Stilt roots — maize, sugarcane · Pneumatophores — Rhizophora.



Stems in Disguise

Potato was a stem in hiding — a **tuber** with "eyes" that are really nodes! Ginger was an underground **rhizome**. Cucumber climbed with **tendrils**, Citrus guarded itself with **thorns**, and Opuntia's flat green stem, a **phylloclade**, did the work of leaves.

> 💡 **REMEMBER**
> **Thorns are modified axillary buds. A phylloclade is a flattened green stem that photosynthesises.**



Leaf Talk

The leaves compared their veins: **reticulate venation** (a network) in dicots and **parallel venation** in monocots. Then they lined up on the stem by **phyllotaxy** — alternate in china rose, opposite in guava and whorled in Alstonia.

💡 **REMEMBER**

Pea leaflets become tendrils; cactus leaves become spines.



Flower Arrangements

Flowers grouped themselves into an **inflorescence**. In **racemose**, the main axis kept on growing; in **cymose**, it ended in a flower. Each flower had four whorls: **calyx**, **corolla**, **androecium** and **gynoecium**.

💡 **REMEMBER**

Hypogynous — superior ovary · Perigynous — half-inferior (plum, rose) · Epigynous — inferior ovary (guava, cucumber).



Folded in the Bud

Bindu peeped into buds to see how petals fold — **aestivation**. Calotropis petals just touched: **valvate**. China rose petals overlapped one way round: **twisted**. Cassia and gulmohur were **imbricate**. Pea had one big standard, two wings and a keel: **vexillary**.

💡 **REMEMBER**

Vexillary (papilionaceous) aestivation is typical of pea and bean.



Where Ovules Sit

Inside the ovary, ovules sat by **placentation**: **marginal** in pea, **axile** in tomato and lemon, **parietal** in mustard, **free central** in Dianthus and **basal** in sunflower and marigold.

💡 **REMEMBER**

Placentation = arrangement of ovules within the ovary.



Fruits and Families

A mango proudly showed it was a **drupe**: a thin epicarp, a fleshy, edible **mesocarp** and a hard, stony endocarp. A coconut nodded — "My mesocarp is fibrous." Then three families waved goodbye: **Fabaceae** (pea family), **Solanaceae** (potato family) and **Liliaceae** (lily family).

💡 **REMEMBER**

Fabaceae — pea, soybean · Solanaceae — potato, tomato, brinjal · Liliaceae — tulip, Aloe, Gloriosa.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Morphology p.1 · roots p.1 · stem p.1 · leaves p.1 · flowers p.1 · fruits p.1 · seeds p.1 · prop roots p.2 · stilt roots p.2 · pneumatophores p.2 · tuber p.3 · rhizome p.3 · tendrils p.3 · thorns p.3 · phylloclade p.3 · reticulate venation p.4 · parallel venation p.4 · phyllotaxy p.4 · inflorescence p.5 · racemose p.5 · cymose p.5 · calyx p.5 · corolla p.5 · androecium p.5 · gynoecium p.5 · aestivation p.6 · valvate p.6 · twisted p.6 · imbricate p.6 · vexillary p.6 · placentation p.7 · marginal p.7 · axile p.7 · parietal p.7 · free central p.7 · basal p.7 · drupe p.8 · mesocarp p.8 · Fabaceae p.8 · Solanaceae p.8 · Liliaceae p.8



QUICK QUIZ

Do you remember?

1. Pneumatophores are found in:

- Rhizophora
- Banyan
- Maize
- Carrot

2. Potato is a modified:

- Root
- Stem
- Leaf
- Fruit

3. Vexillary aestivation is seen in:

- Pea
- China rose
- Calotropis
- Cassia

4. Axile placentation is seen in:

- Pea
- Mustard
- Tomato
- Marigold

5. The edible part of mango is the:

- Epicarp
- Mesocarp
- Endocarp
- Seed

Answers: 1-Rhizophora · 2-Stem · 3-Pea · 4-Tomato · 5-Mesocarp